

Supplemental Article S1

From Fidelity to Semantics: A Taxonomy of Explainable AI Evaluation Metrics and a Paired Empirical Comparison of LIME and SHAP

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Supplemental Article S1

This supplementary document contains six tables referenced from the main text: (S1) the exact evaluation rubric and prompts used in EXP4; (S2) the LIME `kernel_width` sensitivity probe results; (S3) pairwise feature correlations for the Adult Income dataset; (S4) the second-reviewer taxonomy audit template; (S5) the LIME `num_samples` sensitivity probe results; and (S6) the EXP6 masking-scheme sensitivity breakdown.

Table S1 — EXP4 Evaluation Rubric and Judge Prompts

EXP4 employed three LLM judges each scoring the same 192 single-explanation cases on seven semantic dimensions using a structured rubric: 96 from the Adult benchmark (EXP2) and 96 from the cross-dataset cohorts (EXP3; German Credit and Breast Cancer), across four model families and all four explainers. In the original cohort, all three judges scored 147 of these cases, the subset on which its ICC is computed. The template was rendered per instance with the following fixed prompt structure:

System instruction (delivered once per judge session):

You are an expert AI evaluator specialising in Explainable AI (XAI) for tabular data. Your task is to evaluate the quality of natural-language explanation summaries produced for a binary classification model (UCI Adult Income dataset, target: annual income >\$50K). Evaluate each explanation on seven dimensions using a 1–5 integer Likert scale. Return your response in strict JSON format with keys `scores` (object), `rationales` (object), and optional `flags` (array). Do not add commentary outside the JSON block.

Per-instance user prompt (template variables in `{{}}`):

Dataset: UCI Adult Income (Census Income, 14 features).
Model: `{{model_name}}` (e.g., Random Forest, XGBoost).
Explainer: `{{method_name}}` (SHAP or LIME).
Predicted label: `{{prediction_label}}`. **True label:** `{{true_label}}`.
Explanation text:
`{{explanation_text}}`
Rate the explanation on the seven dimensions below (1 = very poor, 5 = excellent). Return strict JSON.

Judge models and configuration:

Table 1: EXP4 rubric: seven semantic evaluation dimensions, operational definitions, and scoring anchors. Temperature = 0.0 for all judges. Three conditions: `hidden_label` (standard), `label_visible` (bias probe), `rubric_alt` (rubric sensitivity). Primary ICC results come from the `hidden_label` condition. The released analysis code averages each case-judge pair over whichever conditions and replicates were run; whether the original run included the two probes cannot now be verified. The replication reports both views.

Dimension	Operational definition	Scoring anchors
Clarity	Understandable organisation and wording for a non-expert reader.	1 = incomprehensible; 3 = requires domain knowledge; 5 = clear to any reader
Concision	Compact relative to useful information content; avoids unnecessary redundancy.	1 = verbose or padded; 3 = adequate; 5 = maximally informative per word
Actionability	Supports recourse, decision review, or explicit next steps for a user.	1 = no actionable content; 3 = vague suggestions; 5 = specific, feasible actions
Audit Usefulness	Supports inspection of model behaviour by an auditor or data scientist.	1 = no audit value; 3 = partially informative about model logic; 5 = fully supports bias/error investigation
Completeness	Provides sufficient local decision information to understand the specific prediction.	1 = missing key drivers; 3 = covers main drivers; 5 = full local decision context
Semantic Plausibility	Coherent with the instance context and general domain knowledge.	1 = implausible or contradictory; 3 = plausible; 5 = highly consistent with domain expectations
Overall Quality	Holistic semantic quality integrating all dimensions above.	1 = very poor; 3 = acceptable; 5 = excellent

- Judge A: `gpt-4o-mini` (OpenAI), temperature 0.0, `max_tokens` 512.
- Judge B: `claude-3-haiku-20240307` (Anthropic), temperature 0.0, `max_tokens` 512.
- Judge C: `gemini-1.5-flash` (Google), temperature 0.0, `max_tokens` 512.

Replication panel (same 192 cases, all three conditions, three replicates each, via OpenRouter; the original models are no longer all available): `openai/gpt-5.4-mini`, `anthropic/claude-haiku-4.5` and `google/gemini-3.8-flash`, temperature 0.0, `max_tokens` 4000 to accommodate reasoning tokens. Its templates were rebuilt from this table and read the dataset from each case record: the case inventory spans three datasets, although the system instruction above names Adult only. The alternative-rubric wording is new, because this table does not record the original. The rebuilt templates, rendered prompts and raw responses are released with the artifacts.

All judges used identical prompts. The three Jinja templates that rendered this rubric under the three conditions (`exp4_semantic_eval_v1.j2`, `_label_visible.j2`, `_alt.j2`) were not retained in the repository; the table above is therefore the authoritative record of the instrument rather than a transcription of a committed file. The unrelated `src/prompts/templates/explanation_eval.j2` present in the repository is an earlier three-dimension instrument from EXP1 and was *not* used for EXP4. The seven dimensions and the 1–5 integer scale are corroborated by the recovered EXP4 scoring schema (`src/evaluation/exp4_schema.py`) and by the per-dimension reliability tables.

Table S2 — LIME `kernel_width` Sensitivity Probe

Table 2 reports LIME metrics under four `kernel_width` settings (RF model, seed 42, $N = 100$, `num_samples`=1000, `num_features`=10). Fidelity and stability are computed only over instances with at least one non-zero weight (“valid instances”).

Table 2: LIME sensitivity to `kernel_width` (RF, seed 42, $N = 100$). Fidelity = Pearson r between $|w_i|$ and per-feature prediction drop. Stability = mean pairwise cosine similarity under $T = 15$ Gaussian perturbations ($\sigma = 0.1$). Reference condition: `kernel_width`=3.0.

<code>kernel_width</code>	Valid instances	Fidelity	Stability	Sparsity	Cost (ms)
0.25	0/100 (degenerate)	—	—	0.000	30
0.75	0/100 (degenerate)	—	—	0.000	30
3.0 (<i>ref.</i>)	96/100	0.518	0.000	0.087	30
10.0	100/100	0.441	0.664	0.093	32

Kernel widths of 0.25 and 0.75 produce zero-weight explanations for all 100 instances, because the exponential neighbourhood collapses in the 103-dimensional one-hot-encoded feature space. The reference value (3.0) is the minimum viable setting. `kernel_width`=10.0 yields stable explanations (0.664) by averaging over a wider neighbourhood, at a cost of lower fidelity (0.441). These results confirm that LIME instability under the reference configuration is not a calibration artifact: narrowing the kernel does not improve stability; it degenerates the explanations. Widening the kernel improves stability but at the cost of locality, so LIME’s near-zero stability at the reference width depends on the configuration.

Table S3 — Adult Income Feature Correlation Structure

Table 3 contextualises the out-of-distribution masking discussion in the main text. Numeric–numeric correlations were computed as Pearson r over the raw (pre-preprocessing) dataset ($n = 48,842$); categorical–categorical associations as Cramér’s V over pairwise-complete cases, dropping records missing either value. `workclass` and `occupation` are unobserved on the same 2,809 records, so admitting missingness as its own category would manufacture an association between them. Only pairs with $|r| > 0.10$ or $V > 0.15$ are shown.

The four strongest associations (`relationship`–`sex`, `marital-status`–`relationship`, `marital-status`–`sex`, `occupation`–`sex`) all involve protected attributes, meaning that independent feature masking during fidelity evaluation discards statistically plausible joint values for these features. The functional duplicate `education`/`education-num` means that any fidelity measurement which masks one but not the other is measuring the effect of partial information removal rather than true feature exclusion. These associations do not invalidate the SHAP–LIME comparison—both methods are evaluated under identical masking protocols—but they limit the interpretation of absolute fidelity values as measures of causal feature importance.

Table S4 — Second-Reviewer Taxonomy Audit

Table 4 reports the independent second-reviewer audit used to stress-test single-reviewer bias in the taxonomy construction. An independent human second reviewer, recruited through the project’s human-evaluation dashboard, audited a stratified subset of 16 papers sampled across the six thematic clusters in the coded corpus: faithfulness/robustness, human-grounded, taxonomy/survey, benchmark/toolkit, LLM-judge, and counterfactual/recourse. For each paper, Reviewer 2 records

Table 3: Selected feature associations in the Adult Income dataset. Pearson r for numeric pairs; Cramér’s V for categorical pairs. `education` and `education-num` are functionally identical (one-to-one mapping; $V \equiv 1.0$). Pearson r over the full pre-split dataset ($n = 48,842$); Cramér’s V over pairwise-complete cases.

Feature 1	Feature 2	Type	Association
<i>Deterministic / near-perfect relationships</i>			
<code>education</code>	<code>education-num</code>	ordinal encode	$V \equiv 1.0$
<i>Strong categorical associations ($V > 0.40$)</i>			
<code>relationship</code>	<code>sex</code>	Cramér’s V	0.647
<code>marital-status</code>	<code>relationship</code>	Cramér’s V	0.488
<code>marital-status</code>	<code>sex</code>	Cramér’s V	0.460
<code>occupation</code>	<code>sex</code>	Cramér’s V	0.435
<i>Moderate categorical associations ($0.15 < V \leq 0.40$)</i>			
<code>workclass</code>	<code>occupation</code>	Cramér’s V	0.216
<code>education</code>	<code>occupation</code>	Cramér’s V	0.196
<code>occupation</code>	<code>relationship</code>	Cramér’s V	0.176
<i>Numeric correlations ($r > 0.10$; all others < 0.10)</i>			
<code>education-num</code>	<code>hours-per-week</code>	Pearson r	0.144
<code>capital-gain</code>	<code>education-num</code>	Pearson r	0.125

whether the source satisfies inclusion criteria and independently assigns the four taxonomy axes. Agreement is then computed against Reviewer 1 coding. For multi-label axes, report both exact percent agreement and Jaccard similarity; for single-label inclusion decisions, report Cohen’s κ .

The row-level audit is archived in `docs/reports/paper_bc/second_reviewer_audit_results.csv`. This subset audit supports codebook refinement and bias detection; a full-corpus independent human second-reviewer audit remains future work before broader inter-rater reliability claims are made.

Table S5 — LIME `num_samples` Sensitivity Probe

Table 5 reports LIME metrics under three `num_samples` settings (RF model, seed 42, $N = 100$, `kernel_width=3.0`, `num_features=10`), testing whether the near-zero stability observed at the reference setting is an under-sampling artifact.

Stability remains near-zero at every sample budget tested and fidelity is essentially flat throughout. Doubling or halving the sample budget neither stabilizes the explanations nor changes the fidelity ordering: under the reference kernel width, more samples do not raise LIME’s stability. Table S2 shows that the kernel width does, so the near-zero reading is a property of the reference configuration rather than of the method.

Provenance and scope. This probe is an earlier exploratory measurement whose output artifact is not preserved, unlike the `kernel_width` probe (Table S2), which re-derives from `outputs/analysis/lime_kernel_width_sensitivity.csv`. Both tables state the same reference condition (RF, seed 42, $N = 100$, `num_samples=1000`, `kernel_width=3.0`) but report different values for it—fidelity 0.461 against 0.518, stability 0.014 against 0.000—so the two probes were run with different instrumentation and do not describe the same measurement. Both are reported for their direction, not as point estimates comparable with each other or with the EXP2 means. The

Table 4: Second-reviewer audit summary. Cohen’s κ is not estimable for the inclusion decision because the audited subset contains included records only.

Audit item	Sample	Agreement metric	Result	Action
Title/abstract inclusion decision	16 stratified records	Percent agreement; Cohen’s κ	100.0%; κ not estimable	Adjudicate include/exclude conflicts
Evaluation target axis	16 included records	Percent agreement; Jaccard similarity	62.5%; mean $J=0.802$	Revise axis definitions if disagreement clusters
Evidence source axis	16 included records	Percent agreement; Jaccard similarity	56.2%; mean $J=0.771$	Clarify proxy, benchmark, human, and LLM labels
Quality property axis	16 included records	Percent agreement; Jaccard similarity	25.0%; mean $J=0.657$	Split or merge ambiguous construct labels
Task context axis	16 included records	Percent agreement; Jaccard similarity	81.2%; mean $J=0.906$	Clarify modality and domain-transfer labels
Overall adjudication log	All disagreements	Count and qualitative summary	12/16 records; mean axis $J=0.784$	Preserve final rationale in audit notes

Table 5: LIME sensitivity to `num_samples` (RF, seed 42, $N = 100$). Stability = mean pairwise cosine similarity under $T = 15$ Gaussian perturbations ($\sigma = 0.1$). Reference condition: `num_samples=1000`. A historical exploratory measurement whose values cannot be re-derived from the released artifacts; see the provenance note below.

<code>num_samples</code>	Fidelity	Stability
500	0.452	0.011
1000 (<i>ref.</i>)	0.461	0.014
2000	0.459	0.016

conclusion they support jointly, and which survives the discrepancy, is directional: more samples do not recover stability, while a wider kernel does, at a cost in fidelity. Re-running the surviving sensitivity script would produce a new measurement rather than a reproduction of this table, because it sweeps a different grid over a different base configuration.

Table S6 — EXP6 Masking-Scheme Sensitivity Breakdown

Table 6 reports the per-scheme breakdown for EXP6, the top-feature masking-sensitivity probe over the Adult RF stratum (the five RF runs at $N = 50$, one per seed, with top-10 stored explanations from EXP2). With five pairs, the exact Wilcoxon test cannot reach $\alpha = 0.05$ under any scheme, so the sign agreement below is descriptive. Four masking schemes are compared: independent zero masking, mean replacement, marginal empirical replacement, and grouped mean replacement (which masks correlated processed feature sets together, e.g. `education/education-num`, `marital-status/relationship/sex`, and `workclass/occupation`).

SHAP’s fidelity advantage direction is stable across all four schemes. Its magnitude, measured as the top- k prediction-drop gap, does not increase, and falls overall, under progressively more correlation-aware masking ($+0.098 \rightarrow +0.098 \rightarrow +0.081 \rightarrow +0.053$), consistent with the feature-correlation discussion in Supplementary Table S3: independent masking of correlated features inflates the apparent SHAP–LIME gap relative to schemes that account for feature dependence. The drop-correlation delta does *not* follow this ordering — it is highest under marginal empirical replacement ($+0.625$) and lowest under grouped mean replacement ($+0.494$) — so the attenuation claim is

Table 6: SHAP–LIME top- k prediction-drop advantage and drop-correlation advantage under four masking schemes (Adult RF stratum, five runs at $N = 50$, one per seed). SHAP’s direction is positive in all five paired seed cells under every scheme.

Masking scheme	Mean SHAP–LIME gap	Mean drop-correlation delta
Independent zero masking	+0.098	+0.515
Mean replacement	+0.098	+0.512
Marginal empirical replacement	+0.081	+0.625
Grouped mean replacement	+0.053	+0.494

supported by the prediction-drop gap alone and is not a general property of every masking-sensitivity endpoint.